The Systems Behind Modern AI

Sam Ly
Professor Thanh Nguyen
CS 4630

October 30, 2026

Abstract

Modern machine learning and AI runs primarily on Graphics Processing Units (GPUs), hardware initially designed to render graphics for video games. We briefly discuss the history of GPU architecture, and the justification behind some key innovations. AI models are composed of neural networks, which rely on the matrix multiplication as a computational primitive. Thus, efficient implementations of matrix multiplication are crucial in modern AI systems. To explore these implementations, we discuss the CUDA programming model, and develop the roofline model as a guiding principle for efficient hardware utilization. The demand for AI-specific compute has led to the development of numerous alternative parallel processing architectures such as Tensor Processing Units, Neural Processing Units, and Wafer Scale Engines. We conclude with a brief discussion of future evolutions of these systems and their justifications.